EXPERIMENT 25

Text Generation Using GPT-3 / OpenAI API

Aim

To generate text from a prompt using an OpenAI language model API.

Procedure

1. Install the OpenAI Python package.
2. Create an API key in the OpenAI platform and store it as an environment variable.
3. Use the SDK to send a prompt to an available text-generation model.
4. Print the generated response.
5. Keep API keys private and check model access and billing.

Program

```
# Install: pip install openai
# Set OPENAI_API_KEY in your environment before running.
from openai import OpenAI

client = OpenAI()
response = client.responses.create(
    model="gpt-3.5-turbo-instruct",
    input="Write three sentences about protecting the environment."
)
print(response.output_text)
```

Output

```
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Example output:
Protecting the environment keeps our air and water clean.
Reducing waste conserves valuable resources.
Planting trees supports wildlife and helps absorb carbon dioxide.
(Note: output is generated by the API and may vary.)
```

Result

Thus, a prompt was sent to an OpenAI text-generation API and the returned text was displayed. Model availability may depend on the account and API.